Condensed Matter Physics I

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Exercise Sheet 4

Exercise 1 — *Planes and directions in crystals* [medium]

Families of lattice planes in crystals are described using Miller indices (hkl). These indices refer to a normal vector perpendicular to the planes (but given in units of the reciprocal lattice) and can be determined via the intersections of a plane with the coordinate axes of the chosen primitive or conventional unit cell.

(a) Consider the families of planes with Miller indices (100), (110), and (111) of a face-centered cubic (fcc) crystal with respect to the conventional cubic unit cell. Draw these planes. What are the indices of these planes when they refer to the primitive axes?

(b) Consider a simple cubic lattice. Determine the spacing between two neighboring planes of the (110) family. Then determine the reciprocal lattice vector $\vec{G}_{110}$ and compare its direction to the direction of the direct-lattice plane normal. What is the orientation of $\vec{G}_{110}$ relative to the (110) planes?

Exercise 2 — *Diffraction and interference in 1D gratings* [medium]

Consider a one-dimensional diffraction grating consisting of N identical slits separated by a distance a.

(a) Consider a periodic grating of slits $N = \infty$ with negligible slit width. How does the diffraction pattern look like?

(b) What happens if one increases the slit width?

(c) Consider again a negligible slit width. What happens if one reduces the number of slits N? For instance to 20, 4, or 2? What does the diffraction pattern look like?

Exercise 3 — *Reciprocal lattice of a 2D Bravais lattice* [medium]

Consider a 2D Bravais lattice spanned by primitive vectors $\vec{a}_1, \vec{a}_2$ with acute angle $\angle(\vec{a}_1, \vec{a}_2) < 90°$. Let $A = (\vec{a}_1 \times \vec{a}_2) \cdot \hat{e}_z$ be the *signed* cell area ($A > 0$ here), and define $A_c = |\vec{a}_1 \times \vec{a}_2|^2 = A^2$. The reciprocal vectors $\vec{b}_1, \vec{b}_2$ are required to satisfy $\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$. For

$$\vec{a}_1 = (0.4,\, 0)\ \text{nm}, \qquad \vec{a}_2 = (0.1,\, 0.2)\ \text{nm},$$

do the following:

(a) Compute the (signed) area $A = (\vec{a}_1 \times \vec{a}_2) \cdot \hat{e}_z$ and $A_c = A^2$.

(b) Compute $\vec{b}_1, \vec{b}_2$ and their magnitudes $|\vec{b}_1|$, $|\vec{b}_2|$.

(c) Verify numerically $\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$.

(d) Sketch the reciprocal lattice points $\vec{G} = h\vec{b}_1 + k\vec{b}_2$ ($h, k \in \mathbb{Z}$) and indicate the *first Brillouin zone* as the Wigner–Seitz cell around $\Gamma = \vec{0}$.

(e) In the *direct* lattice, draw the (11), (10), and (52) planes (lines in 2D).

Exercise 4 — *Structural analysis of copper* [advanced]

Copper has a face-centered cubic (fcc) lattice with one atom per lattice point.

(a) A second-order Bragg reflection from the (001) planes is observed at $2\theta = 54.3°$ using Cu K_α radiation with wavelength $\lambda = 1.5413\,Å$. Use Bragg's law to determine the lattice constant a of copper.

(b) The first-order (001) reflection is not observed. Explain why this reflection is absent in fcc crystals, using only geometrical reasoning.

(c) Consider now the (420) set of planes. Which of the following lattice types exhibit a (420) reflection?

- simple cubic (sc)
  sc structure viewer (TU Graz)
- body-centered cubic (bcc)
  bcc structure viewer (TU Graz)
- face-centered cubic (fcc)
  fcc structure viewer (TU Graz)

Use the structure viewer on the TU Graz webpage to draw the planes and explain your answer again using only geometrical reasoning.